Assignment 1 - Ottimizzazione e simulazione in condizioni di incertezza

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1 Introduction [1]

The Newsvendor problem is a classical stochastic optimization problem arising in inventory management. The objective is to determine the optimal order quantity for a product facing uncertain demand while balancing the costs associated with overstocking and understocking.

In real-world applications, the exact probability distribution of demand is generally unknown and must be estimated using limited historical observations. Consequently, the optimal order quantity is computed using estimated parameters rather than the true ones. Since the Newsvendor solution depends nonlinearly on these parameters through the critical-fractile formula, even moderate estimation errors may lead to suboptimal decisions and a measurable degradation in profit.

In this assignment, a Monte Carlo simulation framework is developed in MATLAB to analyze the impact of uncertainty on the Newsvendor model. The analysis focuses on two main uncertainty sources:

- estimation uncertainty in the demand distribution parameters,
- uncertainty in the salvage value of unsold inventory.

The objective is to quantify how these uncertainties affect the quality of the resulting ordering policy.

The main performance indicator considered is the performance ratio:

$$R = \frac{\Pi(\hat{Q})}{\Pi(Q^*)},$$

where:

- $\Pi(\hat{Q})$ is the realized expected profit obtained using the estimated optimal quantity,
- $\Pi(Q^*)$ is the theoretical optimal expected profit obtained under perfect information.

A value of R close to 1 indicates that the estimated decision performs nearly optimally, while lower values indicate degradation caused by uncertainty.

2 Mathematical Model [3][1]

2.1 Classical Newsvendor Model

Consider a single-period inventory problem characterized by:

- selling price p,
- purchasing cost c,
- salvage / return value s,
- stochastic demand:

$$D \sim \mathcal{N}(\mu, \sigma^2).$$

The underage and overage costs are defined respectively as:

$$C_u = p - c,$$

$$C_o = c - s.$$

The Newsvendor optimality condition is given by the critical ratio:

$$F(Q^*) = \frac{C_u}{C_u + C_o}.$$

Assuming normally distributed demand, the optimal order quantity is:

$$Q^* = \mu + \sigma \Phi^{-1} \left(\frac{C_u}{C_u + C_o} \right),$$

where $\Phi^{-1}(\cdot)$ denotes the inverse cumulative distribution function of the standard normal distribution.

The corresponding expected profit is evaluated analytically using the closed-form expression for normally distributed demand.

2.2 Demand Parameter Estimation

In practice, the true values of μ and σ are unknown. Instead, they are estimated from a finite historical sample:

$$D_1, \ldots, D_n.$$

The estimated parameters are:

$$\hat{\mu}, \qquad \hat{\sigma}.$$

These estimated quantities are then used to compute the order quantity:

$$\hat{Q}.$$

Because the estimation process is affected by sampling noise, the computed quantity is generally different from the true optimum.

2.3 Salvage Value Uncertainty

A second uncertainty source is the salvage value of unsold products.

Let:

$$\hat{s}$$

be the predicted salvage value and:

$$s_{\text{real}}$$

be the actual realized salvage value.

The uncertainty is modeled as:

$$s_{\text{real}} = \hat{s} + \varepsilon,$$

with:

$$\varepsilon \sim \mathcal{N}(0, \tau^2).$$

Increasing τ corresponds to larger uncertainty in the selling price.

3 Monte Carlo Simulation Framework [2]

The numerical experiments are implemented in MATLAB using repeated Monte Carlo simulations.

For each replication, the following steps are performed:

1. The true demand distribution parameters are fixed.
2. A finite historical demand sample is generated.
3. The demand parameters are estimated from the sample.
4. The estimated optimal quantity $\hat{Q}$ is computed using the Newsvendor formula.
5. Future demand and salvage-price realizations are generated.
6. The realized profit associated with $\hat{Q}$ is evaluated.

7. The result is compared with the theoretical optimal expected profit.

The simulation is repeated many times in order to estimate the statistical effect of uncertainty.

The main performance metrics considered are:

3.1 Performance Ratio

$$R = \frac{\Pi(\hat{Q})}{\Pi(Q^*)}.$$

3.2 Regret

$$\text{Regret} = \Pi(Q^*) - \Pi(\hat{Q}).$$

4 Experimental Design

4.1 Experiment A — Effect of Sample Size

The first experiment analyzes the impact of the historical sample size used for parameter estimation.

The considered values are:

$$n \in \{5, 10, 20, 50, 100, 250, 500, 1000\}.$$

For each sample size, the simulation evaluates:

- average performance ratio,
- average regret,

The objective is to understand how additional data improves the reliability of the ordering policy.

4.2 Experiment B — Salvage / Return Price Uncertainty

The second experiment investigates uncertainty in the salvage value.

The standard deviation of the salvage-price error is varied:

$$\tau \in \{0, 0.5, 1, 2, 3, 4, 5\}.$$

The objective is to evaluate how uncertainty in salvage conditions affects the quality of the optimal policy.

4.3 Experiment C — Underage and Overage Cost Ratio

The third experiment studies the effect of the cost structure on the robustness of the Newsvendor solution.

The selling price p, purchasing cost c, and salvage price r are modified in order to generate different ratios between underage and overage costs.

The underage and overage costs are defined as

$$C_u = p - c$$

and

$$C_o = c - r,$$

respectively.

The considered cost ratios are therefore

$$\frac{C_u}{C_o}.$$

The experiment includes cases corresponding to different economic regimes:

- $C_u < C_o$: overstocking is more expensive than understocking,
- $C_u = C_o$: balanced cost structure,
- $C_u > C_o$: understocking is more expensive than overstocking.

The following parameter configurations are analyzed:

p	c	r	$\frac{C_u}{C_o}$
11	10	0	0.1
15	10	0	0.5
20	10	0	1
15	10	8	2.5
20	10	8	5
20	10	9	10

The objective is to investigate how underage to overage asymmetry influences the sensitivity of the Newsvendor policy to estimation errors in the demand distribution parameters.

4.4 Experiment D — Demand Variance

The final experiment analyzes the role of demand variability.

The coefficient of variation is varied according to:

$$CV = \frac{\sigma}{\mu}.$$

Representative values include:

$$0.05,\ 0.10,\ 0.15,\ 0.20,\ 0.30,\ 0.40.$$

The objective is to understand how increasing uncertainty in demand amplifies estimation errors and regret.

5 Results and Discussion

The numerical experiments confirm that uncertainty in model parameters can substantially affect the quality of the Newsvendor solution.

5.1 Effect of Sample Size

The results show that small sample sizes produce highly variable parameter estimates. In particular, estimation errors in the standard deviation strongly affect the tail quantile involved in the Newsvendor solution.

As the sample size increases, the estimates become more stable and the performance ratio progressively approaches 1. The variability of the regret also decreases significantly.

This behavior is consistent with statistical estimation theory: larger datasets reduce sampling noise and therefore improve optimization quality.

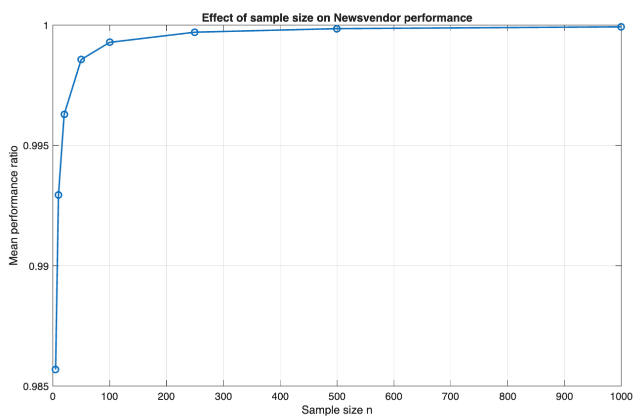

Figure 1: Effect of sample size on the mean performance ratio.

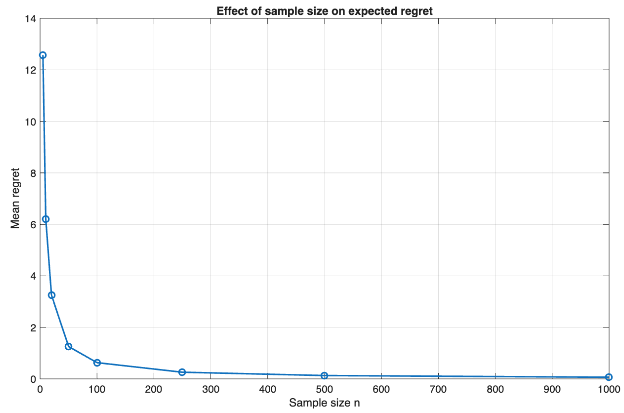

Figure 2: Effect of sample size on the mean regret.

SampleSize	MeanPerformanceRatio	MeanRegret
5	0.985685025041295	12.5779604313178
10	0.992934131787363	6.20847825775648
20	0.996291085646008	3.25886549730132
50	0.99856953305889	1.25689053843946
100	0.999283493624859	0.629563716407652
250	0.999706032433728	0.258296813465247
500	0.999853396988086	0.128813839233652
1000	0.999927807861199	0.063432165820481

Table 1: Table of effects of sample size..

5.2 Effect of Salvage/Return Price Uncertainty

The simulations show that increasing salvage-price uncertainty degrades the quality of the optimal policy.

When the realized salvage value differs substantially from the predicted one, the underage and overage costs change, making the estimated critical fractile and hence optimal order quantity inaccurate. Consequently, the computed order quantity is no longer optimal.

The degradation becomes particularly visible for large values of τ, where both the average regret and performance ratio decreases.

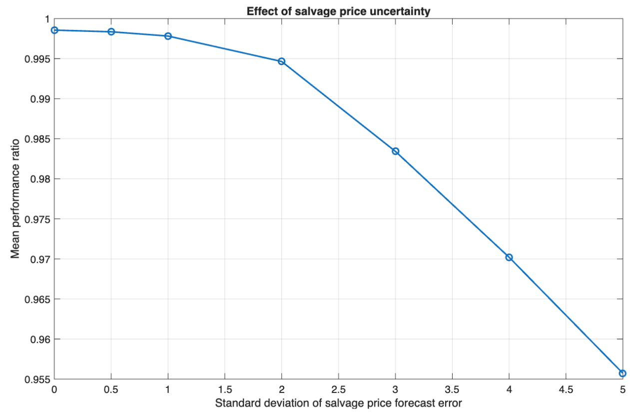

Figure 3: Effect of salvage price uncertainty on the mean performance ratio.

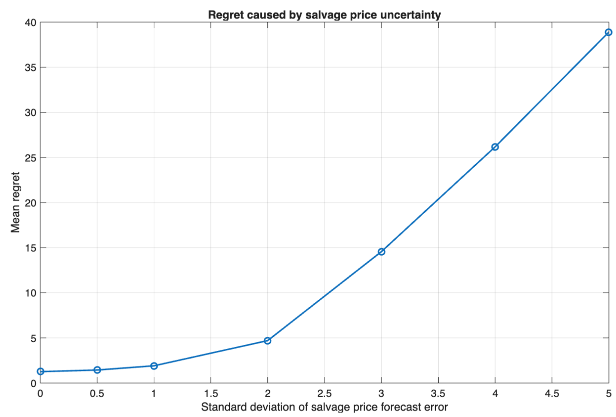

Figure 4: Effect of salvage price uncertainty on the mean regret.

SalvageErrorStd	MeanPerformanceRatio	MeanRegret
0	0.998562301858354	1.26324428718664
0.5	0.998356201982107	1.44433549382987
1	0.997823389364498	1.91249530835494
2	0.994644028416463	4.70606472196623
3	0.983423113309522	14.5654061895832
4	0.970203699114729	26.1807439143764
5	0.955729837547368	38.8983112596087

Table 2: Table of effects of salvage price uncertainty.

5.3 Effect of Cost Asymmetry

The simulations indicate that the sensitivity of the Newsvendor solution strongly depends on the critical fractile.

When underage costs dominate, the optimal solution shifts toward higher inventory levels. In these situations, estimation errors can generate larger financial consequences because excessive ordering may produce significant overstock losses.

Conversely, when overage costs dominate, the policy becomes more conservative and less sensitive to estimation noise.

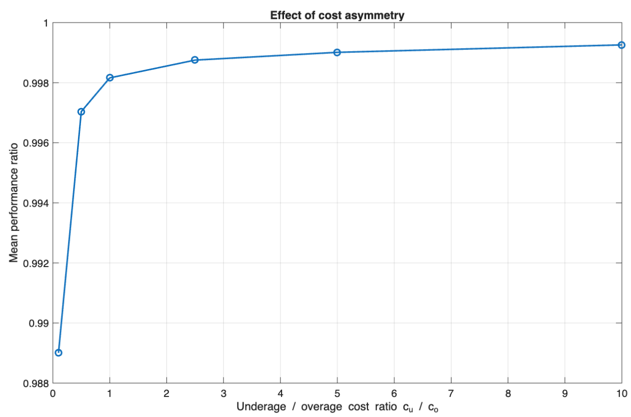

Figure 5: Effect of cost asymmetry on the mean performance ratio.

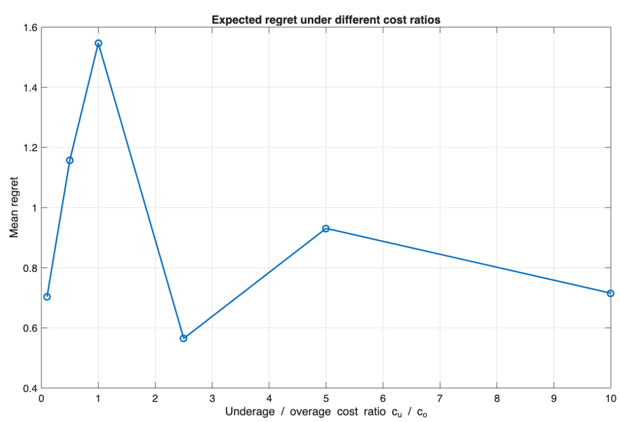

Figure 6: Effect of cost asymmetry on the mean performance regret.

CostRatio	MeanPerformanceRatio	MeanRegret
0.1	0.989007902301239	0.703565363857214
0.5	0.997038305236024	1.1577859173363
1	0.998159524438506	1.54677815448326
2.5	0.998751188439618	0.564978867329055
5	0.999009744076046	0.930875994183608
10	0.999257707521848	0.715574751046743

Table 3: Table of effects of cost asymmetry.

It should be noted that, in order to generate different values of the underage/overage cost ratio, the parameters p, c, and r are modified across the experiments. As a consequence, the scale of the optimal expected profit changes between scenarios.

For this reason, the regret values are not directly comparable across all cost structures, since regret is expressed in absolute profit units and therefore depends on the magnitude of the corresponding optimal profit.

On the other hand, the performance ratio provides a normalized measure of performance degradation and is therefore a more appropriate indicator for comparing robustness across different scenarios.

5.4 Effect of Demand Variance

The numerical results indicate that larger demand variability amplifies the impact of estimation errors.

When the standard deviation is small relative to the mean demand, the Newsvendor solution remains relatively stable. However, for large coefficients of variation, the uncertainty region around the optimal quantity becomes significantly wider.

As a consequence:

- regret distributions become more dispersed,
- performance ratios decrease,
- large degradations become more frequent.

This result is intuitive because higher demand uncertainty makes the inventory decision intrinsically more difficult.

Overall, the experiments illustrate an important concept emphasized during the course: optimization under uncertainty is not only a forecasting problem. Even relatively small estimation errors may propagate nonlinearly through the optimization model and produce significant economic consequences.

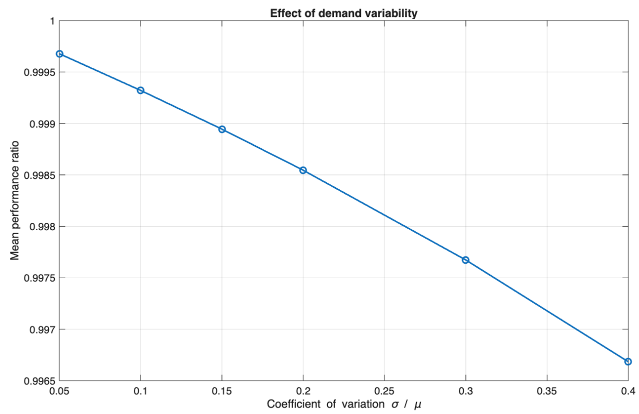

Figure 7: Effect of demand variance / rho on the mean performance ratio.

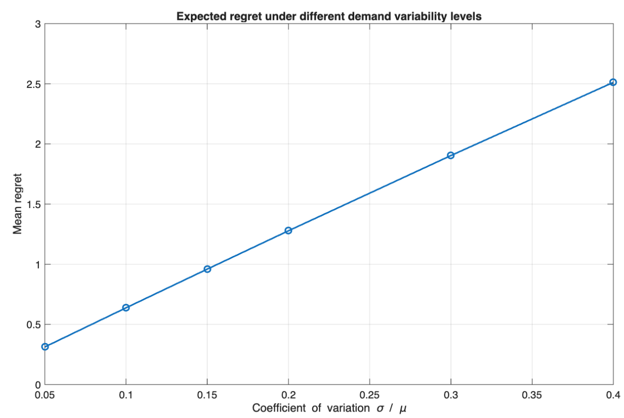

Figure 8: Effect of demand variance / rho on the mean performance regret.

CoefficientOfVariation	MeanPerformanceRatio	MeanRegret
0.05	0.999677807823907	0.312418276871329
0.1	0.999321168547985	0.637645906906987
0.15	0.998944607212821	0.959344805499069
0.2	0.998545323354385	1.27816257741915
0.3	0.99767303083203	1.90342886197505
0.4	0.996683317940966	2.51177322253413

Table 4: Table of effects of demand variance / rho.

6 Conclusion

This assignment presented a Monte Carlo uncertainty analysis of the classical Newsvendor model.

The study investigated the impact of:

- limited historical data,
- salvage-price uncertainty,
- asymmetric cost structures,
- demand variability.

The numerical experiments showed that parameter uncertainty may significantly reduce the performance of the ordering policy compared to the theoretical optimum.

In particular, the simulations highlighted the importance of accurate parameter estimation and the strong interaction between uncertainty and cost asymmetry.

Possible future extensions and shortcomings include:

- Different values of p, c, and r are used to generate different underage/overage cost ratios. Since regret is measured in absolute profit units, its values are not directly comparable across all scenarios. The performance ratio is therefore a more suitable indicator for comparing robustness across different economic regimes.
- In all experiments except the one specifically dedicated to the analysis of sample-size effects, the same historical sample size `hss` $= 50$ is used for parameter estimation. As a possible extension, future analyses could investigate how different historical sample sizes interact with the remaining uncertainty sources and economic regimes considered in the other experiments.
- Future analyses could investigate the combined effect of multiple uncertainty sources simultaneously, rather than studying each factor separately. A more realistic setting could consider the joint interaction between demand estimation errors, salvage-price uncertainty, and different cost structures.
- Another possible improvement would be the introduction of a multi-period or step-by-step ordering simulation. In such a framework, ordering decisions could be updated dynamically over time as new demand information becomes available. This would allow the analysis of adaptive inventory policies and their robustness under uncertainty in a more realistic operational setting.

Disclaimer

LLM tools (namely ChatGPT Auto) were used in an editorial capacity in order to improve the phrasing and presentation of the ideas contained in this report.

Additionally, ChatGPT was used to assist in converting written notes into LaTeX format, which was subsequently compiled using Overleaf.

In the coding section, ChatGPT was also used for the creation and saving of figures and tables within the MATLAB implementation, as well as with parts of the coding syntax and debugging of structural mistakes in the code.

References

[1] P. Brandimarte,
Platonic vs. Real-World Optimization,
Lecture notes for the course
Ottimizzazione e simulazione in condizioni di incertezza,
Politecnico di Torino, 2026.

[2] P. Brandimarte,
Introduction to Stochastic Simulation,
Lecture notes for the course
Ottimizzazione e simulazione in condizioni di incertezza,
Politecnico di Torino, 2026.

[3] J. R. Birge and F. Louveaux, *Introduction to Stochastic Programming*, 2nd Edition, Springer, 2011, pp. 15–17.

A MATLAB Code

A.1 main.m

Listing 1: Main simulation script

```
%% Assignment 1 - Newsvendor Monte Carlo Uncertainty
    Analysis

clear; clc; close all;
addpath('src');

rng(348306);

%% Base parameters

mu = 100;
```

```
sigma = 20;

p = 20;
c = 10;
r = 4;

hss = 50; %historical sample size

n = 10000; %number of simulations

q = optimal_order_quantity(mu, sigma, p, c, r);

ep = expected_profit(q, mu, sigma, p, c, r);

fprintf("True optimal order quantity q* = %.4f\n", q);
fprintf("True optimal expected profit = %.4f\n\n", ep);

%% Experiment 1: Sample size

ss = [5 10 20 50 100 250 500 1000]; %sample sizes

[mean_ratio_sample, mean_regret_sample] =
    run_experiment_sample_size(ss, n, mu, sigma, p, c, r);

figure;
plot(ss, mean_ratio_sample, '-o', 'LineWidth', 1.5);
xlabel('Sample size n');
ylabel('Mean performance ratio');
title('Effect of sample size on Newsvendor performance');
grid on;
saveas(gcf, 'results/figures/sample_size_ratio.png');

figure;
plot(ss, mean_regret_sample, '-o', 'LineWidth', 1.5);
xlabel('Sample size n');
ylabel('Mean regret');
title('Effect of sample size on expected regret');
grid on;
saveas(gcf, 'results/figures/sample_size_regret.png');

%% Experiment 2: Salvage price uncertainty

se = [0 0.5 1 2 3 4 5]; %salvage error

[mean_ratio_salvage, mean_regret_salvage] =
    run_experiment_salvage_uncertainty( ...
    se, hss, n, mu, sigma, p, c, r);

figure;
plot(se, mean_ratio_salvage, '-o', 'LineWidth', 1.5);
```

```
xlabel('Standard deviation of salvage price forecast error')
    ;
ylabel('Mean performance ratio');
title('Effect of salvage price uncertainty');
grid on;
saveas(gcf, 'results/figures/salvage_uncertainty_ratio.png')
    ;

figure;
plot(se, mean_regret_salvage, '-o', 'LineWidth', 1.5);
xlabel('Standard deviation of salvage price forecast error')
    ;
ylabel('Mean regret');
title('Regret caused by salvage price uncertainty');
grid on;
saveas(gcf, 'results/figures/salvage_uncertainty_regret.png'
    );

%% Experiment 3: Cost asymmetry

[cost_ratios, mean_ratio_cost, mean_regret_cost] = ...
    run_experiment_cost_asymmetry(hss, n, mu, sigma);

figure;
plot(cost_ratios, mean_ratio_cost, '-o', 'LineWidth', 1.5);
xlabel('Underage / overage cost ratio c_u / c_o');
ylabel('Mean performance ratio');
title('Effect of cost asymmetry');
grid on;
saveas(gcf, 'results/figures/cost_asymmetry_ratio.png');

figure;
plot(cost_ratios, mean_regret_cost, '-o', 'LineWidth', 1.5);
xlabel('Underage / overage cost ratio c_u / c_o');
ylabel('Mean regret');
title('Expected regret under different cost ratios');
grid on;
saveas(gcf, 'results/figures/cost_asymmetry_regret.png');

disp(cost_ratios)

%% Experiment 4: Coefficient of variation

rho = [0.05 0.10 0.15 0.20 0.30 0.40];

% rho = coefficient of variation

[mean_ratio_rho, mean_regret_rho] = run_experiment_rho( ...
    rho, hss, n, mu, p, c, r);

```

```matlab
figure;
plot(rho, mean_ratio_rho, '-o', 'LineWidth', 1.5);
xlabel('Coefficient of variation \sigma / \mu');
ylabel('Mean performance ratio');
title('Effect of demand variability');
grid on;
saveas(gcf, 'results/figures/cv_ratio.png');

figure;
plot(rho, mean_regret_rho, '-o', 'LineWidth', 1.5);
xlabel('Coefficient of variation \sigma / \mu');
ylabel('Mean regret');
title('Expected regret under different demand variability levels');
grid on;
saveas(gcf, 'results/figures/cv_regret.png');

%% Summary tables

T_sample = table(ss', mean_ratio_sample', mean_regret_sample', ...
    'VariableNames', {'SampleSize', 'MeanPerformance Ratio', 'MeanRegret'});

T_salvage = table(se', mean_ratio_salvage', mean_regret_salvage', ...
    'VariableNames', {'SalvageError', 'MeanPerformance Ratio', 'MeanRegret'});

T_cv = table(rho', mean_ratio_rho', mean_regret_rho', ...
    'VariableNames', {'CoefficientOfVariation', 'MeanPerformance Ratio', 'MeanRegret'});

T_cost = table(cost_ratios', mean_ratio_cost', mean_regret_cost', ...
    'VariableNames', {'CostRatio', 'MeanPerformanceRatio', 'MeanRegret'});

writetable(T_cost, 'results/tables/cost_asymmetry_results.csv');
writetable(T_sample, 'results/tables/sample_size_results.csv');
writetable(T_salvage, 'results/tables/salvage_uncertainty_results.csv');
writetable(T_cv, 'results/tables/cv_results.csv');
```

A.2 src/expected_profit.m

Listing 2: Expected profit function

```
function ep = expected_profit( ...
    q, mu, sigma, p, c, r)

profit_margin = p - c;
co = c - r;

z = (q - mu) / sigma;

loss = standard_loss_function(z);

ep = profit_margin * mu - co * (q - mu) - (profit_margin +
    co) * sigma * loss;

end
```

A.3 src/optimal_order_quantity.m

Listing 3: Optimal order quantity function

```
function q = optimal_order_quantity(mu, sigma, p, c, r)

% mu= mean, sigma=standard dev., p=selling price, c=
    purchasing price,
% r=return price

cu = p - c;
co = c - r;

%cu=cost of underage, co = cost of overage

or = cu / (cu + co);

%q = optimal ratio

q = mu + sigma * norminv(or);

%orq = optimal order quantity

q = max(q,0);

end
```

A.4 src/run_experiment_cost_asymmetry.m

Listing 4: Cost asymmetry experiment

```
function [cost_ratios, mean_performance_ratio, mean_regret] = ...
    run_experiment_cost_asymmetry(hss, n, mu, sigma)

% Cases for each regime:
% cu < co, cu = co, cu > co

p_values = [11 15 20 15 20 20];
c_values = [10 10 10 10 10 10];
r_values = [0  0  0  8  8  9];;

cost_ratios = zeros(size(p_values));
mean_performance_ratio = zeros(size(p_values));
mean_regret = zeros(size(p_values));

for i = 1:length(p_values)

    p = p_values(i);
    c = c_values(i);
    r = r_values(i);

    cu = p - c; % cost of underage
    co = c - r; % cost of overage

    cost_ratios(i) = cu / co;

    oq = optimal_order_quantity(mu, sigma, p, c, r);
    oep = expected_profit(oq, mu, sigma, p, c, r);

    performance_ratios = zeros(n, 1);
    regrets = zeros(n, 1);

    for k = 1:n

        demand_sample = mu + sigma * randn(hss, 1);

        estimated_mu = mean(demand_sample);
        estimated_sigma = std(demand_sample, 1);

        estimated_order_quantity = optimal_order_quantity( ...
            estimated_mu, estimated_sigma, p, c, r);

        estimated_decision_profit = expected_profit( ...
            estimated_order_quantity, mu, sigma, p, c, r);

        performance_ratios(k) = estimated_decision_profit / oep;
        regrets(k) = oep - estimated_decision_profit;

```

```
    end

    mean_performance_ratio(i) = mean(performance_ratios);
    mean_regret(i) = mean(regrets);

end

end
```

A.5 src/run_experiment_rho.m

Listing 5: Rho experiment

```
function [mean_performance_ratio, mean_regret] =
    run_experiment_rho(rho, hss, n, mu, p, c, r)

mean_performance_ratio = zeros(size(rho));
mean_regret = zeros(size(rho));

for i = 1:length(rho)

    rhoi = rho(i);
    true_sigma = rhoi * mu;

    oq = optimal_order_quantity(mu, true_sigma, p, c, r);

    %oq = optimal order quantity

    oep = expected_profit(oq, mu, true_sigma, p, c, r);

    %oep = optimal expected profit

    performance_ratios = zeros(n, 1);

    regrets = zeros(n, 1);

    for k = 1:n

        demand_sample = mu + true_sigma * randn(hss, 1);

        estimated_mean_demand = mean(demand_sample);
        estimated_demand_sigma = std(demand_sample, 1);

        estimated_order_quantity = optimal_order_quantity(
            estimated_mean_demand, estimated_demand_sigma, p,
             c, r);

        estimated_decision_profit = expected_profit(
            estimated_order_quantity, mu, true_sigma, p, c, r
```

```
            );

        performance_ratios(k) = estimated_decision_profit / oep;
        regrets(k) = oep - estimated_decision_profit;

    end

    mean_performance_ratio(i) = mean(performance_ratios);
    mean_regret(i) = mean(regrets);

end

end
```

A.6 src/run_experiment_salvage_uncertainty.m

Listing 6: Salvage price uncertainty experiment

```
function [mean_performance_ratio, mean_regret] = run_experiment_salvage_uncertainty( ...
    se, hss, n, mu, sigma, p, c, r)

q = optimal_order_quantity(mu, sigma, p, c, r); % q = optimal order quantity

oep = expected_profit(q, mu, sigma, p, c, r); %oep = optimal expected profit

mean_performance_ratio = zeros(size(se));
mean_regret = zeros(size(se));

for i = 1:length(se)

    sei = se(i);

    performance_ratios = zeros(n, 1);
    regrets = zeros(n, 1);

    for k = 1:n

        demand_sample = mu + sigma * randn(hss, 1);

        estimated_mu = mean(demand_sample);
        estimated_sigma = std(demand_sample, 1);

        estimated_r = r + sei * randn;
        estimated_r = max(0, min(estimated_r, c - 0.01));

```

```
        estimated_q = optimal_order_quantity(estimated_mu,
            estimated_sigma, p, c, estimated_r);

        estimated_decision_profit = expected_profit(
            estimated_q, mu, sigma, p, c, r);

        performance_ratios(k) = estimated_decision_profit /
            oep;
        regrets(k) = oep - estimated_decision_profit;

    end

    mean_performance_ratio(i) = mean(performance_ratios);
    mean_regret(i) = mean(regrets);

end

end
```

A.7 src/run_experiment_sample_size.m

Listing 7: Sample size experiment

```
function [mean_performance_ratio, mean_regret] =
    run_experiment_sample_size(ss, n, mu, sigma, p, c, r)

q = optimal_order_quantity(mu, sigma, p, c, r);

oep = expected_profit(q, mu, sigma, p, c, r);

%oep = optimal expected profit

mean_performance_ratio = zeros(size(ss));
mean_regret = zeros(size(ss));

for i = 1:length(ss)

    sample_size = ss(i);

    performance_ratios = zeros(n, 1);
    regrets = zeros(n, 1);

    for k = 1:n

        demand_sample = mu + sigma * randn(sample_size, 1);

        estimated_mean_demand = mean(demand_sample);
        estimated_sigma = std(demand_sample, 1);

```

```
        estimated_order_quantity = optimal_order_quantity(
            estimated_mean_demand, estimated_sigma, p, c, r);

        estimated_decision_profit = expected_profit(
            estimated_order_quantity, mu, sigma, p, c, r);

        performance_ratios(k) = estimated_decision_profit /
            oep;
        regrets(k) = oep - estimated_decision_profit;

    end

    mean_performance_ratio(i) = mean(performance_ratios);
    mean_regret(i) = mean(regrets);

end

end
```

A.8 src/standard_loss_function.m

Listing 8: Standard loss function

```
function loss = standard_loss_function(z)

loss = normpdf(z) - z .* (1 - normcdf(z));

end
```